

1. Horizontal Partitioning

RANGE Partitioning

MySQL Workbench

Local Instance MySQL80

File Edit View Query Database Server Tools Scripting Help

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SQL File 10" SQL File 13" range"

```
1 CREATE TABLE STUDENTS_RANGE (
2   id INT NOT NULL,
3   name VARCHAR(50) NOT NULL,
4   age INT NOT NULL,
5   city VARCHAR(50),
6   marks INT
7 )
8 PARTITION BY RANGE (age) (
9   PARTITION p1 VALUES LESS THAN (20),
10  PARTITION p2 VALUES LESS THAN (30),
11  PARTITION p3 VALUES LESS THAN (40),
12  PARTITION p4 VALUES LESS THAN (50)
13 )
```

Result Grid

id	name	age	city	marks
2	Khilan	25	Delhi	90
3	Kaushik	23	Kota	75

SQL Additions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Administration Schemas

PARTITIONS 2 STUDENTS_RANGE 3

Information

No object selected

Action Output

#	Time	Action	Message	Duration / Fetch
1	19:22:56	CREATE TABLE STUDENTS_RANGE (id INT NOT NULL, name VARCHAR(50) NOT NULL, age INT ...	0 row(s) affected	0.047 sec
2	19:22:57	INSERT INTO STUDENTS_RANGE VALUES (1, 'Ramesh', 19, 'Ahmedabad', 85), (2, 'Khilan', 25, 'Delhi', 90), (3, ...	7 row(s) affected Records: 7 Duplicates: 0 Warnings: 0	0.000 sec
3	19:22:57	SELECT PARTITION_NAME, TABLE_ROWS FROM INFORMATION_SCHEMA.PARTITIONS WHERE TABLE...	4 row(s) returned	0.000 sec / 0.000 sec
4	19:22:57	SELECT * FROM STUDENTS_RANGE PARTITION (p2) LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

LIST Partitioning

MySQL Workbench

Local Instance MySQL80

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SQL File 10" SQL File 13" range" list"

```
1 CREATE TABLE STUDENTS_LIST (
2   id INT NOT NULL,
3   name VARCHAR(50) NOT NULL,
4   department VARCHAR(50),
5   department_id INT
6 )
7 PARTITION BY LIST (department_id) (
8   PARTITION p1 VALUES IN (3, 5, 6, 7, 9),
9   PARTITION p2 VALUES IN (13, 15, 16, 17, 20),
10  PARTITION p3 VALUES IN (23, 25, 26, 27, 30),
11  PARTITION p4 VALUES IN (33, 35, 36, 37, 40)
12 )
```

Result Grid

id	name	department	department_id
1	Ramesh	CSE	5

SQL Additions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Administration Schemas

PARTITIONS 1 STUDENTS_LIST 2

Information

No object selected

Action Output

#	Time	Action	Message	Duration / Fetch
1	19:26:04	CREATE TABLE STUDENTS_LIST (id INT NOT NULL, name VARCHAR(50) NOT NULL, department ...	0 row(s) affected	0.063 sec
2	19:26:04	INSERT INTO STUDENTS_LIST VALUES (1, 'Ramesh', 'CSE', 5), (2, 'Khilan', 'MECH', 20), (3, 'Kaushik', 'ECE', ...	7 row(s) affected Records: 7 Duplicates: 0 Warnings: 0	0.000 sec
3	19:26:04	SELECT PARTITION_NAME, TABLE_ROWS FROM INFORMATION_SCHEMA.PARTITIONS WHERE TABLE...	4 row(s) returned	0.000 sec / 0.000 sec
4	19:26:04	SELECT * FROM STUDENTS_LIST PARTITION (p1) LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
5	19:26:04	drop table STUDENTS_LIST	0 row(s) affected	0.016 sec

Object Info Session

HASH Partitioning

MySQL Workbench

Local instance MySQL80

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SQL File 10* SQL File 13* range* list* hash*

Limit to 1000 rows

```
1 CREATE TABLE STUDENTS_HASH (  
2   id INT NOT NULL,  
3   name VARCHAR(50) NOT NULL,  
4   marks INT  
5 )  
6 PARTITION BY HASH(id) PARTITIONS 4;  
7  
8 INSERT INTO STUDENTS_HASH VALUES  
9 (1, 'Ramesh', 85),  
10 (2, 'Khilan', 90),  
11 (3, 'Kaushik', 75),  
12 (4, 'Chaitali', 80);
```

Result Grid

	id	name	marks
1	1	Ramesh	85
5	5	Hardik	88

SQL Additions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Administration Schemas PARTITIONS 1 STUDENTS_HASH 2

Information

No object selected

Action Output

#	Time	Action	Message	Duration / Fetch
1	19:30:16	CREATE TABLE STUDENTS_HASH (id INT NOT NULL, name VARCHAR(50) NOT NULL, marks INT ...	0 row(s) affected	0.047 sec
2	19:30:16	INSERT INTO STUDENTS_HASH VALUES (1, 'Ramesh', 85), (2, 'Khilan', 90), (3, 'Kaushik', 75), (4, 'Chaitali', 80)...	7 row(s) affected Records: 7 Duplicates: 0 Warnings: 0	0.000 sec
3	19:30:16	SELECT PARTITION_NAME, TABLE_ROWS FROM INFORMATION_SCHEMA.PARTITIONS WHERE TABLE...	4 row(s) returned	0.000 sec / 0.000 sec
4	19:30:16	SELECT * FROM STUDENTS_HASH PARTITION (p1) LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

KEY Partitioning

MySQL Workbench

Local instance MySQL80

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SQL File 10* SQL File 13* range* list* hash* key*

Limit to 1000 rows

```
11 (2, 'Khilan', 'khilan@mail.com'),  
12 (3, 'Kaushik', 'kaushik@mail.com'),  
13 (4, 'Chaitali', 'chaitali@mail.com'),  
14 (5, 'Hardik', 'hardik@mail.com'),  
15 (6, 'Komal', 'komal@mail.com'),  
16 (7, 'Huffy', 'huffy@mail.com');  
17  
18 -- Display partition info  
19 SELECT PARTITION_NAME, TABLE_ROWS  
20 FROM INFORMATION_SCHEMA.PARTITIONS  
21 WHERE TABLE_NAME='STUDENTS_KEY';
```

Result Grid

	id	name	email
4	4	Chaitali	chaitali@gmail.com

SQL Additions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Administration Schemas PARTITIONS 3 STUDENTS_KEY 4

Information

No object selected

Action Output

#	Time	Action	Message	Duration / Fetch
1	19:35:42	CREATE TABLE STUDENTS_KEY (id INT NOT NULL, name VARCHAR(50) NOT NULL, email VARC...	0 row(s) affected	0.063 sec
2	19:35:42	INSERT INTO STUDENTS_KEY VALUES (1, 'Ramesh', 'ramesh@gmail.com'), (2, 'Khilan', 'khilan@mail.com'), (3, ...	7 row(s) affected Records: 7 Duplicates: 0 Warnings: 0	0.000 sec
3	19:35:42	SELECT PARTITION_NAME, TABLE_ROWS FROM INFORMATION_SCHEMA.PARTITIONS WHERE TABLE...	4 row(s) returned	0.000 sec / 0.000 sec
4	19:35:42	SELECT * FROM STUDENTS_KEY PARTITION (p1) LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Sub-partitioning

MySQL Workbench

Local instance MySQL80

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Navigation icons

Limit to 1000 rows

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AdministrationSchemas

No object selected

Object InfoSession

SQL File 10"SQL File 13"range"list"hash"keysub"

```
1 CREATE TABLE STUDENT_ORDERS (
2   order_id INT NOT NULL,
3   student_name VARCHAR(50),
4   order_date DATE,
5   order_status VARCHAR(20),
6   order_year INT GENERATED ALWAYS AS (YEAR(order_date)) STORED,
7   order_month INT GENERATED ALWAYS AS (MONTH(order_date)) STORED
8 )
9 PARTITION BY RANGE (order_year)
10 SUBPARTITION BY HASH (order_month)
11 SUBPARTITION (
```

Result Grid

Filter Rows:Export:Wrap Cell Contents

order_id	student_name	order_date	order_status	order_year	order_month
3	Kaushik	2023-01-10	Delivered	2023	1

STUDENT_ORDERS 4

Read OnlyContext HelpSnippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	19:41:29	CREATE TABLE STUDENT_ORDERS (order_id INT NOT NULL, student_name VARCHAR(50), order_...	0 row(s) affected	0.062 sec
2	19:41:29	INSERT INTO STUDENT_ORDERS (order_id, student_name, order_date, order_status) VALUES (1, 'Ramesh', ...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.016 sec
3	19:41:29	SELECT PARTITION_NAME, TABLE_ROWS FROM INFORMATION_SCHEMA.PARTITIONS WHERE TABLE...	6 row(s) returned	0.000 sec / 0.000 sec
4	19:41:29	SELECT * FROM STUDENT_ORDERS PARTITION (p1) LIMIT 0,1000	0 row(s) returned	0.000 sec / 0.000 sec
5	19:42:40	SELECT * FROM STUDENT_ORDERS PARTITION (p2) LIMIT 0,1000	1 row(s) returned	0.000 sec / 0.000 sec

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

SQL script saved to "F:\work\sem7_lab\PR 5\sub.sql"

2. Vertical Partitioning

RANGE COLUMNS Partitioning

The screenshot shows the MySQL Workbench interface with a SQL script for creating a table with range column partitions. The script is as follows:

```
1 CREATE TABLE STUDENT_INVENTORY (  
2     id INT,  
3     student_name VARCHAR(50),  
4     subjects_count INT,  
5     total_marks INT  
6 )  
7 PARTITION BY RANGE COLUMNS(subjects_count, total_marks) (  
8     PARTITION P_low VALUES LESS THAN (3, 100),  
9     PARTITION P_medium VALUES LESS THAN (6, 500),  
10    PARTITION P_high VALUES LESS THAN (10, 1000)  
11 )
```

The output pane shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	19:46:09	CREATE TABLE STUDENT_INVENTORY (id INT, student_name VARCHAR(50), subjects_count INT, ...	0 row(s) affected	0.031 sec
2	19:46:09	INSERT INTO STUDENT_INVENTORY VALUES (1, 'Ramesh', 2, 90), (2, 'Khan', 4, 300), (3, 'Kaushik', 5, 400), ...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec
3	19:46:10	SELECT PARTITION_NAME, TABLE_ROWS FROM INFORMATION_SCHEMA.PARTITIONS WHERE TABLE...	3 row(s) returned	0.000 sec / 0.000 sec
4	19:46:10	SELECT * FROM STUDENT_INVENTORY PARTITION (p_low) LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

The result grid shows the data for the first partition (P_low):

id	student_name	subjects_count	total_marks
1	Ramesh	2	90

LIST COLUMNS Partitioning

The screenshot shows the MySQL Workbench interface with a SQL script for creating a table with list column partitions. The script is as follows:

```
1 CREATE TABLE STUDENT_DEPARTMENT (  
2     id INT,  
3     name VARCHAR(50),  
4     department VARCHAR(50),  
5     admission_date DATE  
6 )  
7 PARTITION BY LIST COLUMNS(department) (  
8     PARTITION p_science VALUES IN ('CSE', 'ECE'),  
9     PARTITION p_arts VALUES IN ('Fashion', 'Hotel Mgmt'),  
10    PARTITION p_management VALUES IN ('IT', 'EEE')  
11 )
```

The output pane shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	19:57:41	CREATE TABLE STUDENT_DEPARTMENT (id INT, name VARCHAR(50), department VARCHAR(50), ...	0 row(s) affected	0.047 sec
2	19:57:41	INSERT INTO STUDENT_DEPARTMENT (id, name, department, admission_date) VALUES (1, 'Ramesh', 'CSE', ...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.016 sec
3	19:57:41	SELECT PARTITION_NAME, TABLE_ROWS FROM INFORMATION_SCHEMA.PARTITIONS WHERE TABLE...	3 row(s) returned	0.000 sec / 0.000 sec
4	19:57:41	SELECT * FROM STUDENT_DEPARTMENT PARTITION (p_arts) LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec
5	19:58:15	SELECT * FROM STUDENT_DEPARTMENT PARTITION (p_science) LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec

The result grid shows the data for the first partition (P_low):

id	name	department	admission_date
1	Ramesh	CSE	2020-01-01
3	Kaushik	ECE	2020-03-01